Business Recommendations - HR Candidate Ranking

Practical deployment guidance from the final executed ranking workflow

Recommendation: use the selected transparent Stage-1 ranker as the default shortlist ordering, exclude connection count from the relevance score, and use explicit recruiter feedback only as a controlled second-pass personalization layer.

1. Business outcome

The project converts a small, repetitive candidate list into a ranked HR shortlist. The 104 source rows represent 53 unique profile groups after duplicate handling, so ranking quality is evaluated at the profile level rather than allowing repeated rows to inflate the evidence.

Business question	Result
How many unique profiles drive the analysis?	53
What is the default model?	Normalized hybrid lexical ranker, Ridge alpha 10
How well does it order the top 10?	NDCG@10 = 0.9628 +/- 0.0621
Does connection count help?	No - best connection model is lower at 0.9562
Can recruiters personalize results?	Yes - 65% Stage 1 + 35% explicit feedback

What changed after the final rerun

The cleaned grouped evaluation selects a hybrid of BM25, word TF-IDF, character TF-IDF, lexical overlap, and HR/intent signals. This replaces the earlier retained-model story and aligns the written recommendations with the final executed notebooks.

2. Recommended operating model

- **Default ranking:** use the selected Stage-1 model to order the general shortlist.
- **Do not reward network size:** connection count reduced grouped-CV NDCG@10 and should stay outside the core relevance score.
- **Recruiter control:** allow explicit stars/rejections to rerank the current search through Rocchio feedback.
- **Keep personalization bounded:** retain the 65% Stage-1 / 35% feedback blend so one interaction does not replace the general ranking.
- **Show rationale:** expose title/query overlap or retrieval evidence where practical so recruiters can understand why a profile surfaced.

3. Evidence behind the recommendation

Model / test	NDCG@10	Business reading
Selected Stage-1 model	0.9628	Best mean grouped-CV ordering in the 192-run screen
Character TF-IDF baseline	0.9535	Strong simple lexical benchmark
BM25 baseline	0.9527	Very similar to character TF-IDF
Word TF-IDF baseline	0.9359	Useful but weaker alone
Best model with connections	0.9562	No benefit from network-size feature

The BM25-versus-word-TF-IDF paired comparison is not decisive: mean difference +0.0168, $p = 0.278$, 95% CI [-0.0161, +0.0498], with 3 wins, 6 ties, and 1 loss across the 10 fold-query observations. The business decision should therefore rely on the combined selected model rather than claiming one retrieval method always wins.

Example shortlist behavior

Stage 1	Profile
1	ID 99 - Seeking Human Resources Position
2	ID 3 - Aspiring Human Resources Professional
3	ID 97 - Aspiring Human Resources Professional
4	ID 6 - Aspiring Human Resources Specialist
5	ID 28 - Seeking Human Resources Opportunities

When source ID 3 is starred in the demonstration, it becomes Stage-2 rank 1, ID 97 becomes rank 2, and ID 99 moves to rank 3. The change is targeted: the base shortlist remains recognizable while profiles using language similar to the explicit preference gain position.

4. Rollout plan

Phase	Action	Success signal
1. Offline readiness	Use the current Stage-1 ranker and fixed query definitions	Notebook outputs and written metrics remain aligned
2. Recruiter pilot	Expose ranked results and optional star/reject controls	Recruiters can identify useful profiles faster
3. Measure behavior	Track shortlist saves, profile opens, and rank positions	Higher engagement near the top without excessive volatility
4. Expand labels	Collect more unique profile judgements and review disagreements	Larger and more stable evaluation set
5. Consider semantic models	Only after stronger labelled coverage exists	Measured gain over the lexical baseline

5. Governance and measurement

Metrics to monitor

- **Ranking quality:** NDCG@10 remains the primary offline metric because it uses the full 0-3 relevance scale.
- **Relevant-result coverage:** MAP@10 and MRR remain useful but currently saturate near 1.0 and are less discriminating.
- **Recruiter behavior:** profile opens, stars, shortlist saves, and time-to-useful-profile should be measured in a pilot.
- **Feedback stability:** monitor how far Stage-2 rankings move after each explicit interaction.
- **Data quality:** continue checking unique IDs, duplicate consistency, grade completeness, and exact preservation of source fields.

Guardrails

- Do not interpret ranking score as a probability of hiring success.
- Do not use connection count as a proxy for candidate quality.
- Keep explicit recruiter feedback auditable and reversible.
- Review location and title text for possible proxy effects before broad deployment.
- Do not infer business impact from offline metrics alone.

6. Decision summary

Proceed with the selected lexical Stage-1 ranker for a controlled recruiter pilot. Keep connection count out of the relevance score. Offer Rocchio feedback as an optional second pass. Prioritize more unique labelled profiles and pilot behavior measurement before increasing model complexity.